DIY Breadboard WIFI-LCD Color Screen Wiring Tutorial for Xiaozhi AI Terminal

1. Hardware Required for DIY

- Development Board: ESP32-S3-DevKitC-1 (select WROOM N16R8 module)
- Digital Microphone: INMP441 or ICS43434 Module
- Power Amplifier: MAX98357A
- Cavity speaker: 8Ω 2~3W or 4Ω 2~3W
- Wires: One box of jumpers, several DuPont wires
- 2 pieces of 400-hole breadboards
- SPI-LCD Color Screen (This article takes the 1.54-inch screen as an example; for other screens, please select and compile the corresponding breadboard wifi-LCD firmware on your own)
- 6*6mm Vertical Tactile Switch (Optional)

In addition, you may also need a multimeter, a soldering iron kit, a pliers set, a Type-C data cable, and a PC for firmware flashing.

Note: Except for the SPI-LCD color screen, please refer to the previous tutorial for the list DIY Hardware List and Wiring Tutorial for Xiaozhi AI Chatbot Breadboard

1.1 SPI-LCD Color Screen (with driver board, soldered 8Pin header)

This tutorial takes a 1.54-inch LCD color screen as an example, and the product parameters are as follows:

- 1.54-inch IPS color screen, supporting RGB 65K color display, with rich display colors
- 240X240 resolution, clear display
- IPS full-view panel, ultra-wide viewing range
- Using a 4-wire SPI serial bus, it only requires 8 IO pins to operate

2. Wiring Scheme for Breadboard WIFI-LCD Color Screen

2.1 Wiring Comparison Table for Development Board and Component Modules

Note: Please wire according to the pins in the table. The wiring photos corresponding to the development board are for reference only.

- Connect the ESP32S3 development board 's **bolded numbers/letters pins** , and microphone, digital power amplifier, LCD display etc. **bolded English letters pins** , **with wires, short-circuit means also connecting wires together** .

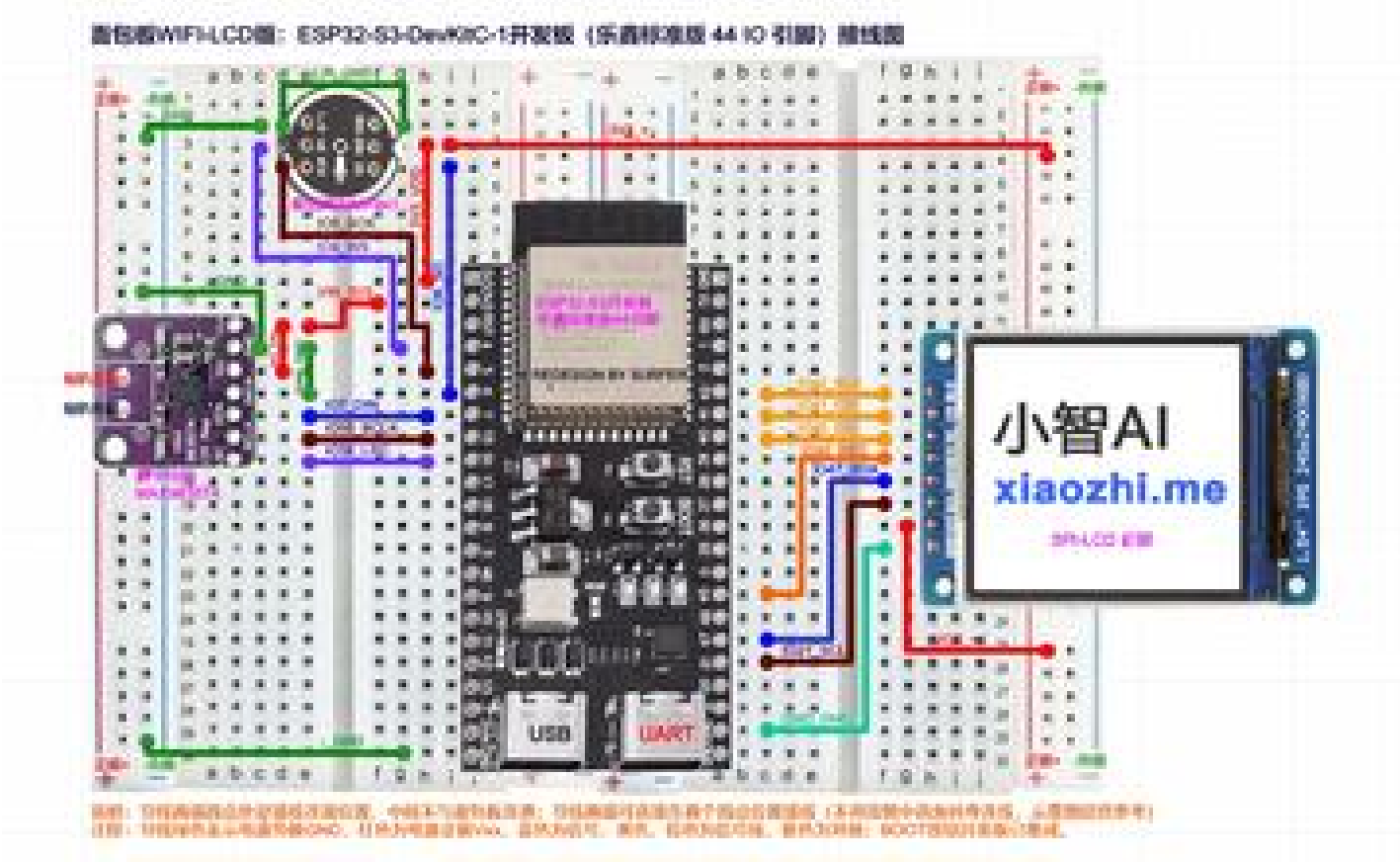
ESP32S3 Development Board	**Microphone INMP441/ICS-43434 (I2S Interface)**
GPIO**4**	**WS** Data Selection
GPIO**5**	**SCK** Data Clock
GPIO**6**	**SD** Data Output
3V3	**VDD** Power Supply Positive 3.3V
GND	**GND** Ground **Short Circuit** **L/R** Left/Right Channel
	Digital Power Amplifier MAX98357A
GPIO**7**	**DIN** Digital Signal
GPIO**15**	**BCLK** Bit Clock
GPIO**16**	**LRC** Left/Right Clock
3V3 /3.3V	**Vin (or VCC) Power Input Short Circuit SD** Shutdown Channel
GND	**GND** Ground **Short** **GAIN** Gain and Channel
	Audio + Connect **to the positive terminal of the speaker** (**usually red wire**, if unsure, ask the seller, use a multimeter to measure, or use other methods)
	Audio - Connect **to Speaker Negative**
	Display (SPI-LCD 8Pin Interface)
GND	**GND** Ground
3V3	**VCC** Positive Power Supply
GPIO**21**	**SCL** Clock Line
GPIO**47**	**SDA** Data Signal
GPIO**45**	**RES** Reset
GPIO**40**	**DC** Data Selection
GPIO**41**	**CS**Chip Select
GPIO**42**	**BLK** Backlight

2.2 ESP32-S3-DevKitC-1 Development Board Version 1 (Espressif Standard Edition 44 IO Pins) LCD Screen Wiring Diagram

This wiring scheme applies to the LCD firmware version >=V1.2.0 of the Xiaozhi AI Chatbot Breadboard WiFi Edition. For easy and quick setup and demonstration, flying wires (jumpers) are used for connection. Please refer to the 2.1 Wiring Connection Table for specific pins.

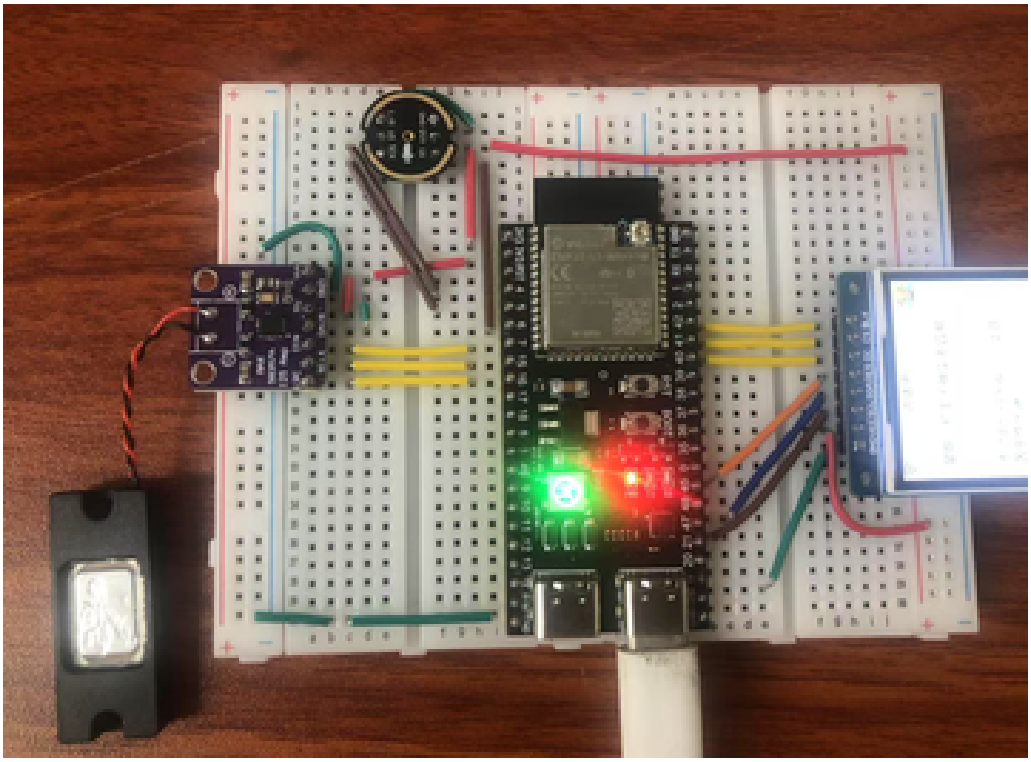
Note: The current firmware does not support the ML307R-4G module for the time being.

- **ESP32-S3 Development Board is a 44 -pin standard breadboard DIY. Please refer to the wiring diagram below:**



Note: If the **MAX98357 digital power amplifier used by the user is in BGA package (the chip is smaller than the one in the figure above) , the short wire GAIN_GND (the green short wire on the left in the figure above) may not be connected. If the user's wiring works properly, there is no need to worry about it.**

- **Please refer to the table for the pin connections of the development board!!! The preview image is for reference only regarding the appearance style!!!**
- After the wiring is completed, a rough preview of the effect is shown in the following figure:

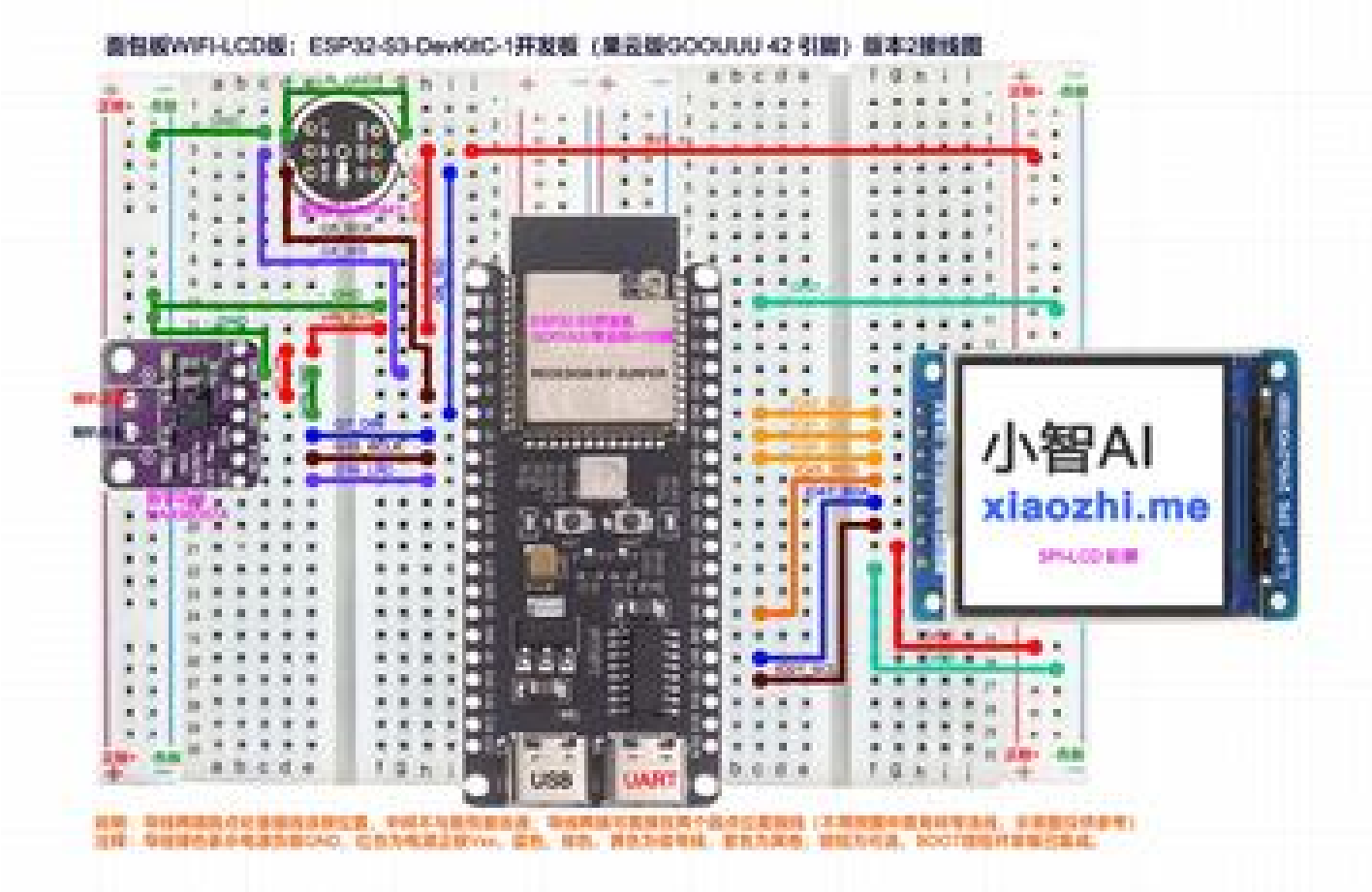


Note: Use short wires to minimize signal interference as much as possible.

2.3 ESP32-S3-DevKitC-1 Development Board Version 2 (GOOUUU Guoyun Edition 42 IO Pins) LCD Screen Wiring Diagram

For wiring the ESP32S3 Guoyun Development Board, please also follow the wiring reference table in 2.1.

- **ESP32-S3 Guoyun 42 Pin Development Board Breadboard DIY, please refer to the wiring diagram below:**



Note: If the **MAX98357 digital power amplifier used by the user is in BGA package (the chip is smaller than the one in the figure above)，the short wire GAIN_GND (the green short wire on the left in the figure above) may not be connected. If the user's wiring works properly, there is no need to worry about it.**

- **Please refer to the table for the pin connections of the development board!!! The preview image is for reference only regarding the appearance style!!!**

Related illustrations to be supplemented

2.4 Jumper Color Scheme Description

In the case of limited jumper styles, try to unify the color scheme as much as possible, and the unified color scheme also makes it easier for others to understand.

For example, VCC uses red, GND uses gray and green, and digital signals use blue, brown, and orange, which can prevent the reverse connection of positive and negative poles.

2.5 FAQ on Common Wiring Issues

1. After burning the firmware, the RGB light does not light up.
   It is necessary to check whether the two solder joints next to the RGB light have been connected. If they are not soldered, you can first connect them with a wire, then restart and check if the light will turn on.
2. How to check for circuit faults?
   - Without connecting the power supply, use a multimeter to test whether the wires are conductive, such as two GND pins that are far apart or two 3.3V pins.
   - After connecting the power supply, check whether the voltage between GND and other pins is within the normal value.

3. Tutorial Bin Firmware

This tutorial uses a 1.54-inch screen with a corresponding resolution of 240x240. If you need the firmware, please download, flash, and test it yourself.

To access the latest version of the firmware, please visit the link (see the last row of the table): https://www.xiaozhi.me/download

v2.0.2_bread-compact-wifi-lcd-240x240.zip
2.23MB

4. Reference link for purchasing related kits

Special Note: The following purchase links are for reference only. Product quality and service are guaranteed by the merchants. If there are any issues, please contact the group administrator for feedback.

Product Name: AI Xiaozhi 1.54-inch Color Screen Material Kit Recommended by Brother Xia, Artificial Intelligence AI Board with esp32-S3-N16R8

Taobao Link:https://item.taobao.com/item.htm?id=993488894351

Reference price: 79 yuan

Community and Communication

QQ Group

Group members are experienced, so if you have any questions, you can ask and communicate in the group.

Welcome to participate in improving this document.